Estimation and Confidence Intervals

8

8.1 INTRODUCTION

For estimating a parameter value, it is important to know (i) a point estimate, (ii) the amount of possible error in the point estimate, that is, an interval likely to contain the parameter value, and (iii) the statement or degree of confidence that the interval contains the parameter value. The knowledge of such information is called *confidence interval* or *interval estimation*.

Point Estimation

A sample statistic (such as $\bar{x}$, s, or $\bar{p}$) that is calculated using sample data to estimate most likely value of the corresponding unknown population parameter (such as μ, σ, or p) is termed as *point estimator*, and the numerical value of the estimator is termed as *point estimate*.

Illustration: For example, if we calculate that 10 per cent of the items in a random sample taken from a day's production are defective, then the result '10 per cent' is a point estimate of the percentage of items in the whole lot that are defective. Thus, until the next sample of items is not drawn and examined, we may proceed on manufacturing with the assumption that any day's production contains 10 per cent defective items.

Confidence Interval Estimation

A confidence interval estimation is an interval of values computed from sample data that is likely to contain the true population parameter value.

The **confidence interval** estimate of a population parameter is obtained by applying the formula:

$$\text{Point estimate} \pm \text{Margin of error}$$

where $\text{Margin of error} = z_c \times \text{Standard error of a particular statistic}$

z_c = critical value of standard normal variable that represents confidence level (probability of being correct) such as 0.90, 0.95, and so on.

8.2 INTERVAL ESTIMATION OF POPULATION MEAN (σ KNOWN)

Suppose the population mean μ is unknown and the true population standard deviation σ is known. Then for a large sample size $n(\geq 30)$, the sample mean $\bar{x}$ is the best point estimator for the population mean μ. Since sampling distribution is approximately normal, it can be used to compute confidence interval of population mean μ as follows:

$$\bar{x} \pm z_{\alpha/2}\, \sigma_{\bar{x}} \quad \text{or} \quad \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

or
$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the z-value representing an area $\alpha/2$ in the right tail of the standard normal probability distribution and $(1 - \alpha)$ is the *level of confidence* as shown in Fig. 8.1.

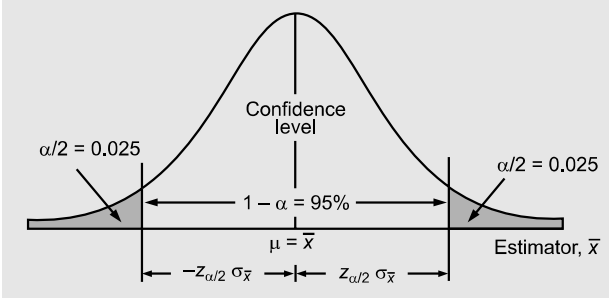


Figure 8.1 Sampling Distribution of Mean x

Illustration 1: For example, if a 95 per cent level of confidence is desired to estimate the mean, then 95 per cent of the area under the normal curve would be divided equally, leaving an area equal to 47.5 per cent between each limit and population mean μ as shown in Fig. 8.2.

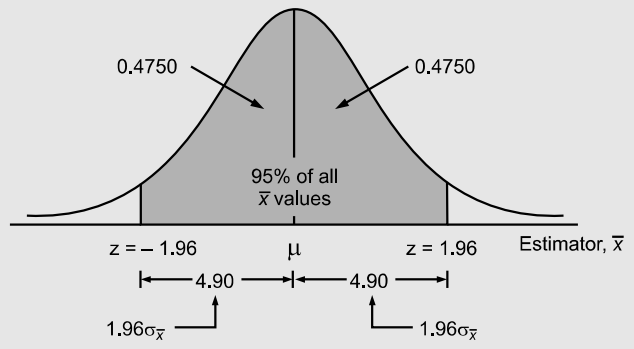


Figure 8.2 Sampling Distribution of Mean $\bar{x}$

Illustration 2: If $n = 100$ and $\sigma = 25$, then $\sigma_{\bar{x}} = \sigma/\sqrt{n} = 25/\sqrt{100} = 2.5$. Using a table of areas for the standard normal probability distribution, 95 per cent of the values of a normally distributed population are within $\pm 1.96\,\sigma_{\bar{x}}$ or 1.96 (2.5) = ± 4.90 range. Hence 95 per cent of the sample means will be within ± 4.90 of the population mean μ. In other words, there is a 0.95 probability that the sample mean will provide a **sampling error** equal to $|\ \bar{x} - \mu\ | = 4.90$ or less. The value 4.90 also provides an upper limit on the sampling error, also called *margin of error*. The value 0.95 is called *confidence coefficient* and the interval estimate $\bar{x} \pm 4.90$ is called a 95 per cent confidence interval.

Example 8.1: The average monthly electricity consumption for a sample of 100 families is 1250 units. Assuming the standard deviation of electric consumption of all families is 150 units, construct a 95 per cent confidence interval estimate of the actual mean electric consumption.

Solution: The information given is $\bar{x} = 1250$, $\sigma = 150$, $n = 100$, and confidence level $(1 - \alpha) = 95$ per cent. Using the 'Standard Normal Curve' we find that the half of 0.95 yields a confidence coefficient $z_{\alpha/2} = 1.96$. Thus confidence limits with $z_{\alpha/2} = \pm 1.96$ for 95 per cent confidence are given by

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1250 \pm 1.96\,\frac{150}{\sqrt{100}} = 1250 \pm 29.40 \text{ units.}$$

Thus, for 95 per cent level of confidence, the population mean μ is likely to fall between 1220.60 units and 1274.40 units, that is, $1220.60 \le \mu \le 1274.40$.

Example 8.2: A survey conducted by a shopping mall group showed that a family in a metro-city spends an average of Rs. 500 on clothes every month. Suppose a sample of 81 families resulted in a sample mean of Rs. 540 per month and a sample standard deviation of Rs. 150, develop a 95 per cent confidence interval estimator of the mean amount spent per month by a family

Solution: The information given is $\bar{x} = 540$, $\sigma = 150$, $n = 81$, and $z_{\alpha/2} = 1.96$ for 95 per cent confidence level. Thus, confidence limits with $z_{\alpha/2} = \pm 1.96$ are given by

$$\bar{x} \pm z_{\alpha/2}\,\sigma_{\bar{x}} = \bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 540 \pm 1.96\,\frac{150}{\sqrt{81}}$$

$$= 540 \pm 40.67 \text{ or Rs. } 499.33 \text{ and Rs. } 580.67.$$

Hence for 95 per cent level of confidence, the population mean μ is likely to fall between Rs. 499.33 and Rs. 580.67, that is, $499.33 \le \mu \le 580.67$.

Example 8.3: The quality control manager at a factory manufacturing light bulbs is interested to estimate the average life of a large shipment of light bulbs. The standard deviation is known to be 100 hours. A random sample of 50 light bulbs gave a sample average life of 350 hours.

(a) Set up a 95 per cent confidence interval estimate of the true average life of light bulbs in the shipment.

(b) Does the population of light bulb life have to be normally distributed? Explain.

Solution: The following information is given:

$$\bar{x} = 350,\ \sigma = 100,\ n = 50, \text{ and confidence level } (1 - \alpha) = 95 \text{ per cent}$$

(a) Using the 'Standard Normal Curve', we have $z_{\alpha/2} = \pm 1.96$ for 95 per cent confidence level. Thus confidence limits are given by

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 350 \pm 1.96 \frac{100}{\sqrt{50}} = 350 \pm 27.72.$$

Hence for 95 per cent level of confidence the population mean μ is likely to fall between 322.28 hours to 377.72 hours, that is, $322.28 \le \mu \le 377.72$.

(b) No, since σ is known and $n = 50$, from the central limit theorem we may assume that $\bar{x}$ is normally distributed.

Interval Estimation for Difference of Two Means

If all possible samples of large size n_1 and n_2 are drawn from two different populations, then sampling distribution of the difference between two means $\bar{x}_1$ and $\bar{x}_2$ is approximately normal with mean $(\mu_1 - \mu_2)$ and standard deviation

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

For a desired confidence level, the confidence interval limits for the population mean $(\mu_1 - \mu_2)$ are given by

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\ \sigma_{\bar{x}_1 - \bar{x}_2}.$$

Example 8.4: The strength of the wire produced by company A has a mean of 4500 kg and a standard deviation of 200 kg. Company B has a mean of 4000 kg and a standard deviation of 300 kg. A sample of 50 wires of company A and 100 wires of company B are selected at random for testing the strength. Find 99 per cent confidence limits on the difference in the average strength of the populations of wires produced by the two companies.

Solution: The following information is given:

Company A: $\bar{x}_1 = 4500,\quad \sigma_1 = 200,\quad n_1 = 50$

Company B: $\bar{x}_2 = 4000,\quad \sigma_2 = 300,\quad n_2 = 100$

Therefore, $\bar{x}_1 - \bar{x}_2 = 4500 - 4000 = 500$ and $z_{\alpha/2} = 2.576$ and

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{40{,}000}{50} + \frac{90{,}000}{100}} = 41.23.$$

The required 99 per cent confidence interval limits are given by

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\ \sigma_{\bar{x}_1 - \bar{x}_2} = 500 \pm 2.576\ (41.23) = 500 \pm 106.20.$$

Hence, the 99 per cent confidence limits on the difference in the average strength of wires produced by the two companies are likely to fall in the interval $393.80 \le \mu \le 606.20$.

8.3 INTERVAL ESTIMATION OF POPULATION MEAN (σ UNKNOWN)

If the standard deviation σ of a population is not known, then it can be approximated by the sample standard deviation s when the sample size n (≥ 30) is large. So, the interval estimator of a population mean μ for a large sample $n(\ge 30)$ with confidence coefficient $1 - \alpha$ is given by

$$\bar{x} \pm z_{\alpha/2}\ s_{\bar{x}} = \bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}.$$